Rein Persistence Theory of Ebola Virus (RPT-EBOV)

ADVANCED THEORETICAL FRAMEWORK & STRUCTURAL DYNAMICS

Independent Analytical Specification | Multi-Stage Entry, Host Interventions, Condensate Dynamics & Latency

1. OVERVIEW & STAGED VIRAL ENTRY SUB-MODEL

This formulation expands standard single-rate within-host dynamics into a multi-step entry pipeline. Target cell infection resolves into sequential proteolytic cleavage, intracellular receptor binding, and allosteric membrane fusion priming:

$$dB/dt = \beta_{eff} \cdot V_W \cdot T - (k_c + d_B) \cdot B$$
$$dK/dt = k_c \cdot B - (k_f + d_B) \cdot K$$
$$dE_W/dt = k_f \cdot K - k_W \cdot E_W$$

Where B denotes bound endosomal cells, K is protease-cleaved NPC1-engaged states, k_c is the cathepsin cleavage rate, and $k_f = \varphi_{variant} \cdot k_{f,0}$ represents fusion priming. Effective entry is defined as $\beta_{eff} = \beta_W \cdot (1 - \varepsilon_{Ab})(1 - \varepsilon_{drug}(D))$. Accelerated fusion variants ($\varphi_{variant} \geq 1$) raise the basic reproductive ratio R_0 and elevate the critical entry blocking threshold required for extinction (e.g., from 88.8% at reference up to 97.2% for 4-fold entry acceleration).

2. HOST-TARGETED PHARMACOLOGICAL TERMS

Small-molecule entry inhibitors acting on cellular trafficking (e.g., TPC2 channels) affect wild-type (V_W) and therapeutic interfering particles (V_T) symmetrically via a Hill response function:

$$\varepsilon_{drug}(D) = D^h / (IC_{50}{}^h + D^h)$$

Reaching the baseline 89% entry reduction benchmark requires $D \approx 8.1 \times IC_{50}$ for $h = 1.0$, and $D \approx 2.8 \times IC_{50}$ for cooperative binding ($h = 2.0$).

3. CONDENSATE & PRIMED CELL DYNAMICS

Viral condensates (NP/VP35 inclusion bodies) partition host mRNAs and viral genomes based on RNA structure, altering interferon-stimulated gene (ISG) signaling:

$$dE_{Co}/dt = \beta_W \cdot (1 - \pi) \cdot P \cdot V_W - k_{Co} \cdot E_{Co}$$
$$dC/dt = \&dots; + g_{cond} \cdot (I_W + I_{Co}) \cdot ISG_{factor}(\pi)$$

Where $\pi \in [0, 1]$ parameterizes interferon protection of primed cells P against wild-type superinfection ($\pi = 0$ recovers baseline).

4. COUNTER-REGULATORY CYTOKINE SUBSYSTEM & LATENT RESERVOIRS

To prevent unrestrained inflammatory escalation, a counter-regulatory mediator state R (representing tissue-protective axes like IL-10/IL-22) balances pro-inflammatory cytokines C. Additionally, a non-productive latent compartment L accounts for quiescent reservoirs:

$$dC/dt = [\rho_I \cdot (I_W + I_{Co}) + \rho_V \cdot V_W + \rho_T \cdot V_T + g_{cond} \cdot I_{total}] / (1 + R/K_R) - d_C \cdot (C - C_0)$$
$$dR/dt = \sigma_R \cdot C / (K_\sigma + C) - d_R \cdot R \quad | \quad dL/dt = v \cdot (E_W + I_W) - (\eta + d_L) \cdot L$$

Extinction is formally partitioned into **Productive Extinction** (active viremia $V_W < V_{ext}$) and **Sterilizing Extinction** (both active viremia $V_W < V_{ext}$ and latent pool $L < L_{ext}$ are cleared). Because TIPs require active co-infection to replicate, quiescent cells L remain inaccessible during latency, requiring structured multi-dose schedules.

5. MODEL EXTENSIONS PARAMETER REFERENCE

Symbol	Parameter Description	Default / Nominal Range
$\varphi_{variant}$	Glycoprotein variant entry fusion multiplier (e.g., A82V)	≥ 1.0 (baseline = 1.0)
k_c, k_f	Cathepsin cleavage and NPC1-engaged fusion priming rates	Fitted constants (day^{-1})
IC_{50}, h	Half-maximal drug concentration and Hill coefficient	51.7 μM (Laudanosine), h ∈ [0.5, 2.0]
π	Interferon protection factor on TIP-primed cells	0.0 – 1.0 (sensitivity range)
g_{cond}	Condensate-driven ISG cytokine amplification gain	Dimensionless coupling factor
R, K_R	Counter-regulatory mediator state and suppression constant	Dynamic state, K_R fitted
L, η, v	Latent pool, reactivation rate, and latency seeding rate	Dynamic state, η & v fitted sensitivity

Theoretical Hypotheses & Falsification Criteria

- **Hypothesis H5 (Cytokine Balancing):** Regulatory arm R suppresses peak pro-inflammatory cytokines C and alters safety boundary trade-offs. *Falsified if: Profile rankings remain unchanged across sweeps of K_R and σ_R.*
- **Hypothesis H6 (Drug Selectivity):** Host-targeted entry inhibitors reduce the required TIP dose for extinction only when wild-type entry suppression exceeds TIP entry suppression. *Falsified if required TIP dose rises monotonically for all $\varepsilon_{drug} > 0$ under symmetric reduction.*
- **Falsifier F6 (Latent Pool Accessibility):** If realistic latent parameters (υ, η) render sterilizing extinction unreachable, therapeutic outcomes must be reclassified from "viral clearance" to "long-term viral suppression."